

Bond Duration

Bond prices are sensitive to interest rate changes, and bond duration is a measure of just how sensitive. For instance, in Exhibit 7.1, an increase in interest rates for the simple bond from 2 percent to 3 percent caused the bond's price to fall by 8.5 percent. This bond has a duration of 8.5, meaning that a 1 percent rise in interest rates leads to an 8.5 percent drop in price. The bond duration is measured in years, and the weighted-dollar average for the time when the cash flows are received in this example is 8.5 years.

Exhibit 7.2 uses the same basic setup as Exhibit 7.1 to provide more insight about how bond prices relate to interest rates and time to maturity. Again, the basic scenario is that we have purchased a bond with a \$1,000 face value and 2 percent annual coupon payments. The current interest rate for a comparable bond is also 2 percent. Sometime shortly after purchase, interest rates change, with the exhibit showing new potential interest rates from 1 to 5 percent. The exhibit shows the new price that bonds with different maturities could sell for after the rate change. The bottom section of the exhibit shows the percentage change in price resulting from the rate change. Here we can clearly see how bond prices move counter to interest rates, and how price fluctuations are more dramatic for longer-term bonds, demonstrating their higher duration. At the extreme, the thirty-year bond would experience a capital *gain* of 25.8 percent if interest rates *fell* by 1 percent, and a 19.6 percent capital *loss* if interest rates *rose* by 1 percent. Price risk increases with time to maturity. If interest rates rose to 5 percent, the capital loss for a thirty-year bond would be 46.1 percent—comparable to a major stock market drop. Despite their reputation as reliable and predictable, bonds can be risky.

Exhibit 7.2

Basic Prices and Interest Rates

Current Bond Value: \$1,000

Coupon Rate: 2% (annual)

Current Interest Rate: 2%

Years to Maturity	New Interest Rate								
	1.0%	1.5%	2.0%	2.5%	3.0%	3.5%	4.0%	4.5%	5.0%
1	\$1,009.90	\$1,004.93	\$1,000.00	\$995.12	\$990.29	\$985.51	\$980.77	\$976.08	\$971.43